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PHOTO
The CARA DC-1 unit on a dashboard
`photos/um_title.jpg`

CARA Driver Companion

Model DC-1

USER MANUAL

Installation • Operation • Maintenance • Disposal

Document no.:	EPD/UM/2024-28/G07
Revision:	1.0
Applies to device software:	0.2.0 and later
Applies to app:	0.3.0 and later
Date of issue:	06-11-2026

⚠ **Read this manual, especially Chapter 1, before using the product. Keep it for future reference.**

Notices and Document Information

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CARA Driver Companion was developed as a course project in *Electronic Product Development (Sessional)*. It is an **educational prototype**: it has not been certified for commercial sale, and it is a driver aid only. It does not drive, brake or steer the vehicle, and it does not replace the driver's own attention to the road and the road signs.

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Revision History

Rev.	Date	Change	Applies to
1.0	06-11-2026	First release, with the DC-1 prototype	Device software 0.2.0; app 0.3.0

Related Documents

Document	Number
Quick Start Guide (one page)	EPD/QSG/2024-28/G07
Product Design Report	EPD/PDR/2024-28/G07
Product Test Report	EPD/PTR/2024-28/G07

Open-Source Components

The product runs open-source software. Each component stays under its own licence; the licence texts are available on request from the support contact in Chapter 8.

Component	Used for	Licence
Raspberry Pi OS, Linux, BlueZ	Operating system and Bluetooth	GPL-2.0 and others
Eclipse Mosquitto	Internal message board between modules	EPL-2.0 / EDL-1.0
LiteRT (TensorFlow Lite)	Running the road-sign model	Apache-2.0
Piper; voice *en_GB-cori-medium*	Natural speech	MIT; see voice model card
eSpeak NG	Back-up speech	GPL-3.0
llama.cpp; Qwen2.5-0.5B-Instruct	On-device trip remark	MIT; Apache-2.0
osmdroid; OpenStreetMap data	Trip map in the app	Apache-2.0; ODbL
IBM Plex Sans and Mono	App fonts	SIL OFL 1.1

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About This Manual

Who Should Read It

This manual is written for **drivers** who use CARA Driver Companion in a car, and for the person who sets it up for them. No electronics knowledge is needed. Sections marked *Technician only* are for trained service staff.

How the Manual Is Organised

Chapter	Use it when you want to …
1 Safety Information	… use the product without harm to people or property
2 Product Overview	… learn what is in the box, the parts, the app and the specifications
3 Installation and Setup	… power, place and pair the product for the first time
4 Operation	… use the product on every drive
5 Maintenance	… clean, update or service the product
6 Troubleshooting	… find the cause of a problem and fix it
7 Disposal and Environment	… recycle or dispose of the product
8 Warranty, Support, Compliance	… get help or check the product's status

Conventions Used

Symbol	Signal word	Meaning
⚠	**DANGER**	Hazard that *will* cause death or serious injury if not avoided.
⚠	**WARNING**	Hazard that *can* cause death or serious injury if not avoided.
⚠	**CAUTION**	Hazard that can cause minor or moderate injury.
—	**NOTICE**	Risk of damage to the product or data; no personal injury.
ⓘ	**TIP / NOTE**	Useful information for better results.
Pair	Sans-serif text	Names of buttons and screen text in the app.
🔊 "*Text*"	Quoted italics	Words the unit speaks.
1., 2., 3.	Numbered steps	Do the steps in the given order.
①	Callout number	Refers to a numbered part in Figure 2.1.

Safety Information

1.1 Intended Use

CARA Driver Companion is a driver aid for private cars. A camera reads road signs ahead and the unit speaks them aloud; it warns when the car is over the last speed limit read; it announces chosen phone messages and calls; and it speaks a short summary at the end of each drive. It is used inside the passenger compartment by adult drivers, together with an Android phone. Use it only as described in this manual.

⚠ WARNING

Do not rely on the product to drive safely. It can miss a sign, read a sign wrongly, or stay silent (for example in rain, at night, in glare or when the camera is blocked). The road signs, traffic rules and the traffic around you always come first. Do not use the product as a speed limiter, for professional or commercial driving, for vehicles other than cars, or for any safety-critical purpose. Misuse can cause accidents and voids the warranty.

1.2 Driving Safety

⚠ WARNING

Distraction can cause a crash. Set up the app, the volume and the announced apps *before* you start driving. Do not hold, look at or operate the phone or the unit while the vehicle is moving. Using a hand-held phone while driving is an offence in India and in many other countries.

- Place the unit so that it does not block your view of the road, the mirrors or the instruments (Section 3.3).
- Never place the unit on or in front of an airbag cover. An airbag can throw it at high speed.
- Make sure the unit and its cable cannot slide, fall or get caught in the pedals or the steering wheel during braking or turning.
- Keep the volume loud enough to hear alerts, but not so loud that you cannot hear horns or emergency sirens.
- If an alert is unclear, keep driving safely and look at the actual sign. Do not stop suddenly to check the unit.

1.3 General Safety Instructions

- Read all instructions before use and keep this manual with the product.
- Use only the supplied cable and a power source that meets Section 3.4.
- Do not use the product if it is damaged, has been dropped, smells of burning or shows signs of overheating.

- Keep the product out of reach of small children and pets.
- Do not modify the product or open the case; there are no user-serviceable parts inside.

1.4 Electrical Safety

⚠ CAUTION

The product runs on low voltage (5 V DC) only. Never connect it directly to the car's 12 V wiring, to a mains socket, or to any adapter other than a USB power source. Do not use the product with wet hands, and keep drinks away from it.

- Use a USB power source rated 5 V DC and at least 2.4 A (power bank) or 2.5 A (car charger).
- Do not use a damaged cable; replace it with a good-quality USB-A to micro-USB cable.
- Route the cable so it is not pinched by doors, seats or the glovebox.

1.5 Power Bank Safety

⚠ WARNING

The power bank (not supplied) contains a lithium battery. It can overheat, burst or catch fire if crushed, punctured, short-circuited or left in a hot car. Use a power bank with a recognised safety mark, do not leave it on the dashboard in the sun, and stop using it if it swells, leaks or gets hot.

1.6 Heat, Environment and Handling

⚠ CAUTION

The case and the air vents ⑥ become warm during use. Do not cover the unit, and keep the air vents clear.

NOTICE

A parked car in the sun can reach 70 °C or more. Do not leave the unit or the power bank in a parked car in direct sunlight. Keep the product away from water, dust and strong magnetic fields. Static electricity can damage the electronics; do not touch the port ⑤ with bare metal.

1.7 Privacy and Data

NOTICE

The unit works without the internet. It does *not* record video or store camera images. It keeps the last 20 trip summaries, with their routes, inside the unit and sends them only to your paired phone. Message and call text is spoken and then discarded; it is never saved in the unit. Only phones paired with the spoken code can connect. When you announce messages, other people in the car can hear them: switch Read message text off or use Quiet mode when you have passengers.

1.8 Symbols on the Product

Table 1.1: Symbols on the product and packaging

Symbol	Meaning	Reference
	Read the user manual before use	ISO 7010 M002
	Warning: hot surface	ISO 7010 W017
	Separate collection of electrical and electronic equipment (do not put in household waste)	WEEE Directive 2012/19/EU
	Direct current input	IEC 60417-5031
	For use inside the vehicle (indoor) only	IEC 60417-5957

Chapter 2

Product Overview

2.1 Description and Key Features

CARA Driver Companion is a small dashboard unit that watches the road through a camera and **speaks** what the driver needs to know, so the driver's eyes stay on the road. It works with a free Android app. Key features:

- **Road-sign alerts.** Reads 20 Indian road signs (IRC:67), such as speed limits, stop, school ahead and speed breaker, and speaks each one once, for example "*Speed limit 50.*"
- **Overspeed warnings.** Compares the car's speed with the last speed limit read and says "*Slow down. Speed limit 50.*" when you are more than 5 km/h over it.
- **Phone announcements.** Reads out the sender of messages from apps you choose (for example WhatsApp), text messages and incoming calls, so you do not have to look at the phone.
- **Trip summary.** When you park, it speaks a short summary of the drive, with one remark written by a small on-device AI. The trip, with its route on a map, appears in the app.
- **Private by design.** Everything runs inside the unit, without the internet. No video is recorded.
- **Natural voice.** A clear, natural female voice; three volume levels.

2.2 What Is in the Box

Table 2.1: Package contents

No.	Item	Qty	Check
1	CARA Driver Companion DC-1 unit (camera and speaker built in)	1	☐
2	USB-A to micro-USB power cable	1	☐
3	Quick Start Guide	1	☐
4	Full User Manual (this document)	1	☐

You also need (not supplied): an Android phone with Android 8.0 or newer, and a USB power bank of 5 V, 2.4 A or more (or a car USB charger, see Section 3.4). If any item is missing or damaged, contact support (Chapter 8) before using the product.

2.3 Parts and Controls

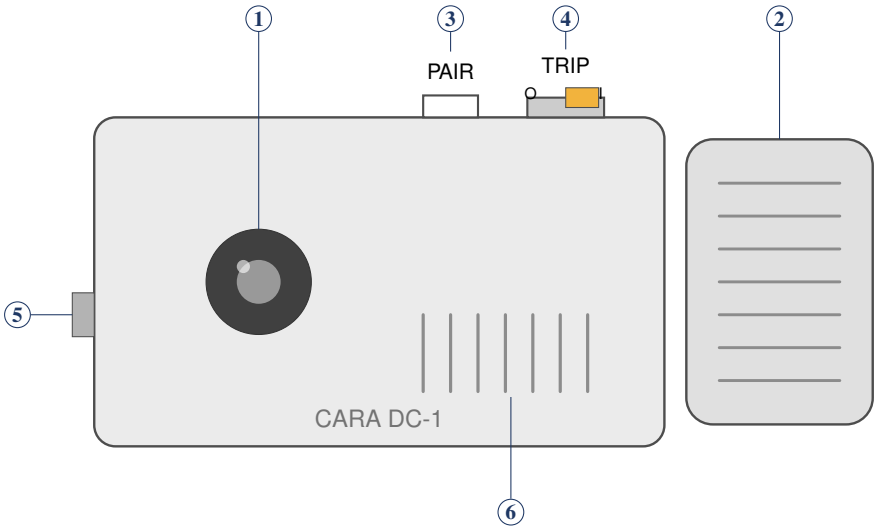


Figure 2.1: Front view: parts and controls

Table 2.2: Parts and controls (see Figure 2.1)

No.	Part	Function
①	Camera window	Looks at the road ahead to read signs. Keep it clean and pointed forward
②	Speaker	Speaks all alerts, announcements and summaries
③	PAIR button	Hold 5 s to let a new phone pair (Section 3.6)
④	TRIP switch	I (on) when you set off, O (off) when you park. Starts and ends a trip
⑤	Power input	Micro-USB socket, 5 V DC. The unit switches on when power is connected
⑥	Air vents	Let heat out. Never cover them

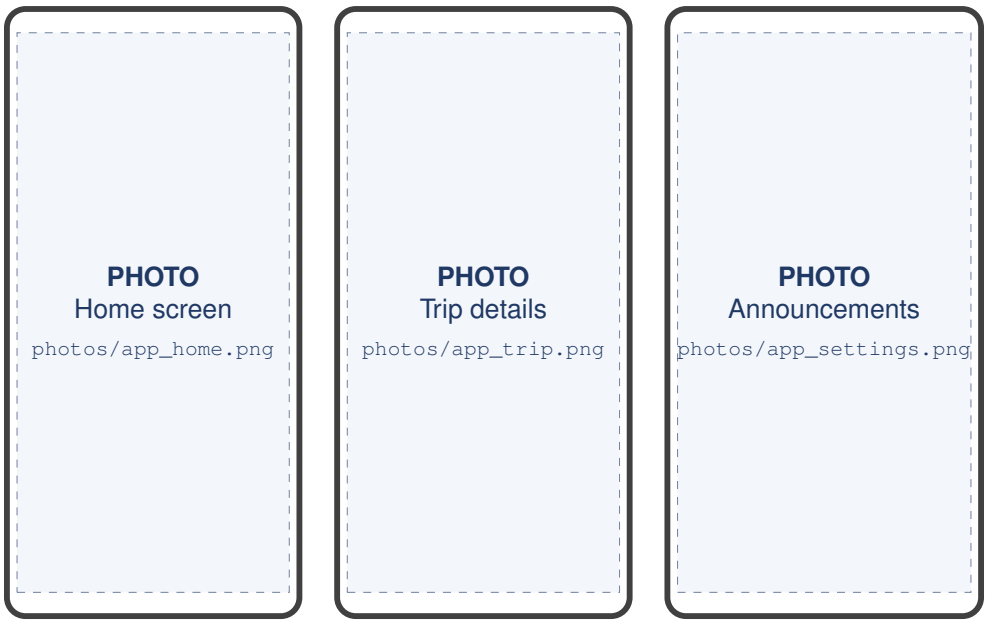
PHOTO
Unit, front
`photos/um_front.jpg`

PHOTO
Unit, top: PAIR and TRIP
`photos/um_top.jpg`

Figure 2.2: The DC-1 unit (photographs)

2.4 The Driver Companion App

The app connects the phone to the unit over Bluetooth. It sends the phone's notifications and its GPS speed to the unit, lets you choose what is announced, and keeps every trip. It has three screens.



(a) Home (b) Trip details (c) Announcements

Figure 2.3: The Driver Companion app

- **Home:** the connection status (a coloured dot and a word), the Phone GPS line, the buttons Send test alert, Disconnect and Pair new, and your three most recent trips.
- **Trip details:** the route on a map, the distance, time and top speed, the spoken summary and the details of the drive.
- **Announcements:** volume, message reading, quiet mode, calls, text messages and the apps to announce (Section 4.9).

2.5 System Block Diagram

The following diagram shows how the parts of the product work together. It is provided for understanding only; it is not needed for normal use.

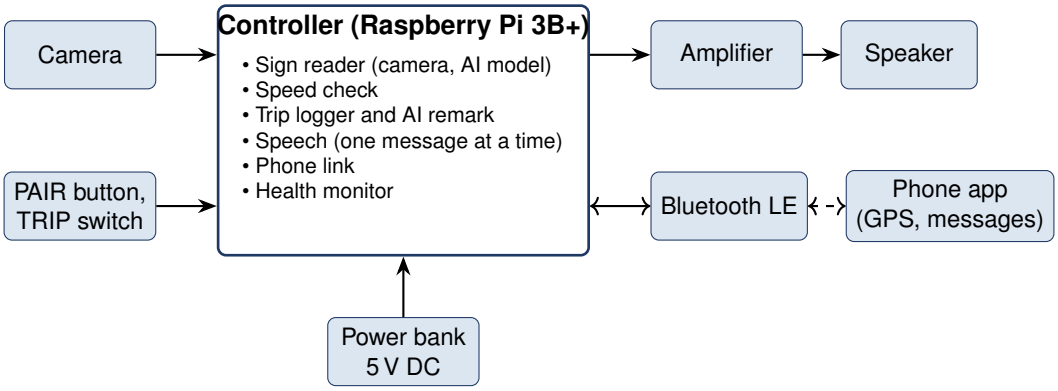


Figure 2.4: Simplified block diagram

2.6 Technical Specifications

Table 2.3: Technical specifications

Parameter	Value
Model	DC-1
Controller	Raspberry Pi 3B+ (quad-core 1.4 GHz, 1 GB RAM)
Power input	5 V DC, micro-USB; source of at least 2.4 A
Power consumption	About 2–5 W, depending on load (controller figures)
Running time on a power bank	About 6–8 h on a 10 000 mAh power bank (estimate)
Ready after power-on	About 60 s, confirmed by "*Self test passed. System ready.*"
Camera	5 MP camera module; signs read at 640×480, 15 frames per second (10 when hot)
Road signs	20 classes from IRC:67: speed limits 20, 30, 40, 50, 60, 80, 100 km/h; stop; give way; no entry; no overtaking; no U-turn; no honking; no parking; school ahead; pedestrian crossing; speed breaker; left curve; right curve; men at work
Sign confirmation	Same sign seen in 3 of 5 frames; about 60 ms processing per frame; speech starts at once
Overspeed warning	More than 5 km/h over the last limit read, for 3 s; repeated at most every 30 s
Speed source	Phone GPS (fixes worse than ± 30 m are ignored)
Audio	3 W, 4 Ω speaker; class-D digital amplifier; three volume levels
Voice	Natural British-English female voice, generated in the unit (offline)
Wireless	Bluetooth Low Energy, 2.4 GHz; pairing with a 6-digit spoken code
Phone app	Android 8.0 or newer
Trip storage	Unit: last 20 trips. App: last 200 trips. Route point about every 25 m
Operating conditions	0–45 °C; inside the vehicle; non-condensing
Storage conditions	-10 to +60 °C; dry
Ingress protection	IP20 (IEC 60529); not protected against water
Dimensions / mass	Main case and speaker pod: see Product Design Report (case limit $125 \times 95 \times 55$ mm, 250 g)
Device software	0.2.0 (app 0.3.0)

Installation and Setup

3.1 Before You Start

- Check the package contents (Section 2.2).
- Charge the power bank fully.
- Have ready: your Android phone, with Bluetooth and Location switched on, and an internet connection to download the app.
- Set up the unit while the car is **parked**.

Table 3.1: Placement requirements

Item	Requirement
Surface	Flat part of the dashboard, near the centre of the windscreen, on a non-slip mat
View	Camera window ① facing straight ahead, with a clear view through the windscreen, inside the area cleared by the wipers
Driver's view	Below your line of sight to the road; not in front of the instruments or mirrors
Airbags	Not on, or in front of, any airbag cover (marked *SRS* or *AIRBAG*)
Clearance	At least 2 cm free space around the air vents ⑥
Phone	Within 5 m of the unit (inside the car), in a holder or a pocket; Bluetooth on

3.2 Unpacking and Inspection

1. Remove the unit and the cable from the packaging.
2. Check the case, the camera window and the cable for cracks, loose parts or bent pins.
3. Keep the packaging for transport and return (Chapter 7).

3.3 Placing the Unit

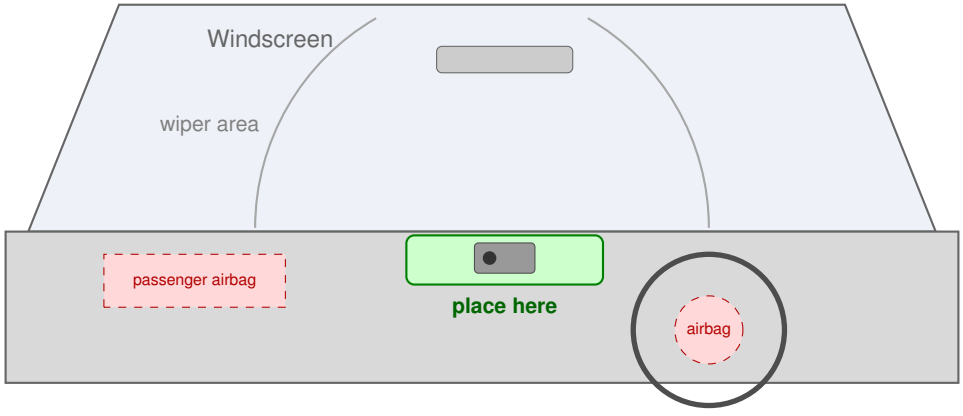


Figure 3.1: Where to place the unit (driver's view, right-hand-drive car)

1. Put a non-slip mat on the centre of the dashboard, away from the airbag covers (Figure 3.1).
2. Place the unit on the mat with the camera window ① facing straight ahead through the windscreen.
3. Check from the driver's seat that the unit is below your view of the road and does not reflect in the windscreen.
4. Check that the camera's view is not blocked by the rear-view mirror, a sticker or a tinted band.
5. Push gently on the unit: it must not slide. If it does, use a stronger non-slip mat.

TIP

The camera reads signs best when it is level and looks slightly to the left of the lane, where Indian road signs stand. A clean windscreen makes a large difference at night and in the rain.

3.4 Connecting Power

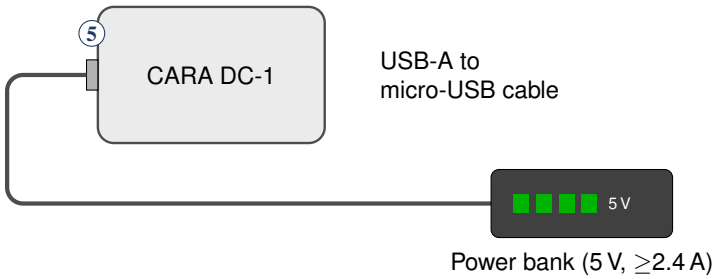


Figure 3.2: Connecting the power bank

Standard: power bank

1. Plug the micro-USB end of the cable into the power input ⑤.
2. Plug the USB-A end into the power bank. The unit switches on.
3. Wait about 60 s. The unit speaks "*Self test passed. System ready.*"
4. Put the power bank where it cannot slide (for example in the door pocket or the centre console), out of direct sunlight.

Option: car USB charger

You can power the unit from a USB charger in the car's 12 V socket, if the charger gives **5 V and at least 2.5 A** on one port.

NOTICE

In most cars the 12 V socket switches off with the key. The unit then loses power at the moment you park, before it can speak the trip summary and send the trip to the app. The trip is closed later and marked *ended with a power cut*. Use a power bank if you want the summary every time. A weak charger makes the unit slow down; if the unit often restarts or alerts come late, change to a stronger charger or a power bank.

3.5 Installing the App

1. On the phone, open the app link from the Quick Start Guide or from `https://ahmed12-stack.github.io/driver-companion_EPD/` and download the Driver Companion app (file `.apk`).
2. Open the downloaded file. If Android asks, allow installing apps from this source.
3. If *Play Protect* shows a warning, tap More details, then Install anyway. (The prototype app is not yet in the Play Store.)
4. Open the Driver Companion app.

Allowing the permissions

Under Permissions needed, tap Allow for each line until the panel shows All permissions allowed.

Table 3.2: App permissions

Permission	Why it is needed
Bluetooth	To find and connect to the unit
Notification access	To announce messages, text messages and calls. Android opens a settings page: switch on *Driver Companion*
Location (precise)	To send the car's speed and the route of each trip to the unit
Notifications	To show the connection status while the app runs in the background

TIP

If Android says *Restricted setting* for Notification access: open **Settings** → **Apps** → **Driver Companion**, tap the ⋮ menu, choose Allow restricted settings, and try again. To keep the connection when the phone sleeps, set **Settings** → **Apps** → **Driver Companion** → **Battery** to Unrestricted.

3.6 Pairing the Phone (First Time)

1. Check that the unit is on and has said "*System ready.*"
2. Press and hold the PAIR button ③ for 5 s, until the unit says 🔊 "*Pairing mode on.*" Release the button.
3. Within 2 minutes, tap Pair in the app.
4. Android shows a pairing box. Type the 6-digit code that the unit speaks, and tap Pair.
5. The unit says 🔊 "*Phone paired.*" The app shows ● Connected.

NOTICE

Pairing mode closes by itself after 2 minutes ("*Pairing mode off.*"). Only a phone that typed the spoken code can connect. Do not pair while someone outside the car can hear the code.

3.7 First-Time Check

1. In the app, tap Send test alert. The unit says a test message.
2. Under Announcements, set the Volume (Section 4.9) and tap Send test alert again until the level suits you.

3. Ask someone to send you a WhatsApp message or a text message. The unit announces it.
4. Check that the Phone GPS line on the home screen turns green (this needs an outdoor view of the sky).

After the first pairing, the app connects again by itself whenever the unit is on and the phone is in range.

Operation

4.1 Switching On and Off

Table 4.1: Power control

Action	How
Switch on	Connect power to ⑤. Ready in about 60 s: "*Self test passed. System ready.*"
Switch off	Set TRIP ④ to O, wait until the trip summary has been spoken and the trip appears in the app, then disconnect the power
Restart	Disconnect the power, wait 10 s, connect it again (use only if the unit does not respond)

NOTICE

The unit is designed to survive a sudden loss of power without damage to its data. A power cut during a drive can lose up to the last minute of that trip.

4.2 What the Unit Says

The unit has no screen and no lights: it tells you its state by voice. The app's status dot shows the connection.

Table 4.2: Status and system messages

The unit says	Meaning / action
"*Self test passed. System ready.*"	Started correctly; ready to drive
"*Self test: . . .*"	Started, but something is wrong (it names the part). See Table 6.2
"*Pairing mode on.*" / "*Pairing mode off.*"	A new phone can pair for 2 minutes / pairing closed
"*Phone paired.*"	A phone has paired successfully
"*Sign alerts paused.*"	The camera has been blocked, dark or dazzled for 10 s. Overspeed warnings continue
"*Sign alerts resumed.*"	The camera can see again
"*Camera not found. Sign alerts are off.*"	The camera is not working. See Table 6.2
"*Warning: the . . . stopped working.*"	One part of the software stopped; it restarts itself within seconds

Table 4.3: Connection status in the app

Dot	Text	Meaning
●	Connected	Phone and unit are linked
●	Searching. . . , Connecting. . .	The app is looking for the unit
●	Bluetooth is off, Disconnected	Switch on Bluetooth, or bring the phone near the unit
●	Not connected	Not paired yet, or you tapped Disconnect

4.3 A Normal Drive

1. Connect the power and wait for "*System ready.*" The app connects by itself.
2. Check that the app shows Connected and a green Phone GPS line. Put the phone away.
3. Set the TRIP switch ④ to I as you set off. The trip starts after 1 s.
4. Drive. The unit speaks road signs, overspeed warnings and your chosen announcements.
5. When you have parked, set the TRIP switch to O. After 5 s the unit speaks the trip summary.
6. Look at the trip in the app later (Section 4.7).

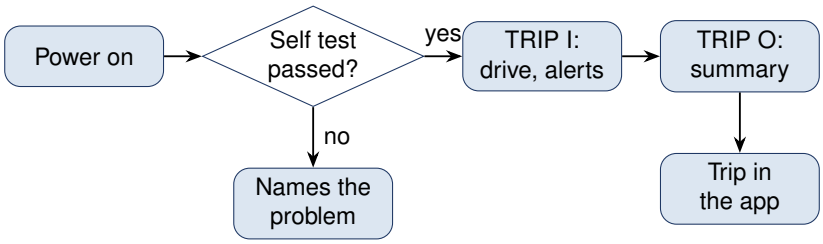


Figure 4.1: Start-up and drive sequence

TIP

Trips shorter than 100 m are not kept (for example when you only move the car in the driveway). Waiting in traffic does not end a trip: only the TRIP switch does.

4.4 Road-Sign Alerts

While a trip is running, the camera looks for the 20 signs listed in Table 4.4. When it has seen the same sign in 3 of 5 pictures, it speaks it once. A sign is spoken again only when it is seen again later.

Table 4.4: Road signs and what the unit says

Sign	The unit says	Sign	The unit says
Speed limit 20–100	"*Speed limit 50.*" (the number read)	No parking	"*No parking.*"
Stop	"*Stop sign ahead.*"	School ahead	"*School ahead.*"
Give way	"*Give way.*"	Pedestrian crossing	"*Pedestrian crossing ahead.*"
No entry	"*No entry.*"	Speed breaker	"*Speed breaker ahead.*"
No overtaking	"*No overtaking.*"	Left curve	"*Left curve ahead.*"
No U-turn	"*No U-turn.*"	Right curve	"*Right curve ahead.*"
No honking	"*No honking.*"	Men at work	"*Men at work ahead.*"

WARNING

Signs can be missed or misread when they are dirty, damaged, bent, hidden by trees or vehicles, very far away, or at night, in rain, fog or strong glare. Other signs are not recognised. Always look at the signs yourself.

4.5 Overspeed Warnings

The unit compares the car's speed (from the phone's GPS) with the **last speed-limit sign it read**.

- When you are more than **5 km/h over** that limit for **3 s**, it says "*Slow down. Speed limit 50.*" This warning is spoken before anything else.
- While you stay over the limit, the warning is repeated at most every 30 s.
- The warning ends when you have been back under the limit for 2 s.
- The limit is forgotten after **5 km**, or when the trip ends, so an old sign never causes a warning later.

NOTICE

No warning is given when no speed-limit sign has been read in the last 5 km, or when the phone's GPS is weak (worse than ±30 m, for example in tunnels or under flyovers). The phone's speed can lag by a second or two.

4.6 Phone Announcements

When the phone is connected, the unit announces:

Table 4.5: Announcements

Event	The unit says (example)
Message from a chosen app	"*WhatsApp message from Rahul.*" With Read message text on: followed by the message
Text message (SMS)	"*Text message from Mum.*"
Incoming call	"*Incoming call from Priya.*"

The unit speaks one thing at a time, most urgent first: overspeed warnings, then road signs, then calls, then messages, then system messages. An announcement that has waited too long is dropped, so you never hear an old call. Quiet mode stops all phone announcements; road alerts continue.

4.7 Trip Summary and Trip History

When you set the TRIP switch to O, the unit speaks:

1. a sentence with the facts of the drive (distance, time, warnings), at once; then
2. one short remark written by the on-device AI, after about 5–8 s. The very first remark after the unit is switched on can take up to a minute.

The numbers always come from the trip log, never from the AI. If the AI's remark does not pass the unit's checks, only the facts are spoken.

In the app, the trip appears under Recent trips; tap it to see:

- the route on a map (drag and pinch to move; the map needs mobile data, the route line does not);
- the distance, trip time and top speed, the average speed, driving time and stops;
- the summary;
- the details: overspeed warnings, time over the limit, road signs read and stops. Harsh braking and Harsh acceleration show Not measured yet on this model.

The unit keeps the last 20 trips and sends each one until the app confirms it, so a trip is not lost when the phone was away. The app keeps the last 200 trips.

4.8 Phone GPS

While connected, the app sends the phone's location and speed to the unit once a second. The Phone GPS line on the home screen shows what is sent: green with the accuracy and speed, or amber when there is no fix yet or the fix is too weak. Keep Location on, with *precise location* allowed.

4.9 Settings Reference

All settings are in the app under Announcements. The unit receives each change at once and keeps it.

Table 4.6: User-adjustable settings

Setting	Options	Default	Description
Volume	Low, Medium, High	Medium	Loudness of everything the unit says
Read message text	On, Off	Off	Off: only the app and the sender are spoken
Quiet mode	On, Off	Off	Road alerts only; no messages or calls
Incoming calls	On, Off	On	Announce who is calling
SMS	On, Off	On	Announce text messages from the default messages app
Apps to announce	WhatsApp, WhatsApp Business, Telegram, Instagram, Messenger, Signal, Gmail	WhatsApp, WhatsApp Business	Tap an app to switch it on (amber dot) or off

Chapter 5

Maintenance

> ⚠ **CAUTION**
>
> Disconnect the power before cleaning or any maintenance work.

5.1 Maintenance Schedule

Table 5.1: Preventive maintenance schedule

Interval	Task	Section	Who
Before each drive	Check the cable, the case and the camera window; listen for "*System ready.*"	—	User
Weekly	Clean the camera window and the windscreen in front of it	5.2	User
Monthly	Clean the case and the air vents; check that the unit does not slide	5.2	User
When announced	Install the new app version or device software	5.4	User / Tech.

5.2 Cleaning

1. Wipe the camera window ① with a dry microfibre cloth, such as a spectacle cloth.
2. Wipe the case with a soft, slightly damp, lint-free cloth. Dry it at once.
3. Clean the air vents ⑥ and the speaker grille ② with a soft, dry brush.

> **NOTICE**
>
> Do not use solvents, glass cleaner, abrasive cleaners or sprays on the unit, and do not let liquid enter the vents or the port.

5.3 Heat

On hot days the unit slows its camera to protect itself, and sign alerts may come slightly later. Keep the vents clear and out of direct sun where possible, and do not leave the unit in a parked car in the sun.

5.4 Software Updates

App

Download the new version from the same link (Section 3.5) and install it over the old one. Your trips, settings and pairing are kept.

Device software (Technician only)

The device software is updated with a laptop, by a technician. Settings, trips, the phone pairing and the voice are kept.

1. Connect a laptop to the unit through the service connection described in the service notes.
2. Copy the new software to the unit and run the installer (`sudo bash install.sh`).
3. Restart the unit if the installer asks.
4. Check that the unit says "*Self test passed. System ready.*" and that the app shows the new software version under the device name.

NOTICE

Do not disconnect the power during an update. If an update stops, run the installer again.

5.5 Storage and Transport

Store the unit in its packaging at −10 to +60 °C, in a dry place. Disconnect the power bank and store it separately, about half charged.

5.6 Spare Parts and Service

Table 5.2: Replaceable parts

Part	Description / specification	Who replaces it
Power cable	USB-A to micro-USB, rated 2.4 A or more	User
Camera module	Raspberry Pi camera, 5 MP, ribbon cable	Technician
Speaker	3 W, 4 Ω, 3-inch	Technician
Amplifier	MAX98357A I2S class-D amplifier board	Technician
microSD card	Card with the device software	Technician

Every module plugs in; no soldering is needed to replace it. Service must be carried out only by the CARA team or a trained technician. Contact: `cara.support.dc1@gmail.com`.

Troubleshooting

6.1 Quick Diagnosis Flow

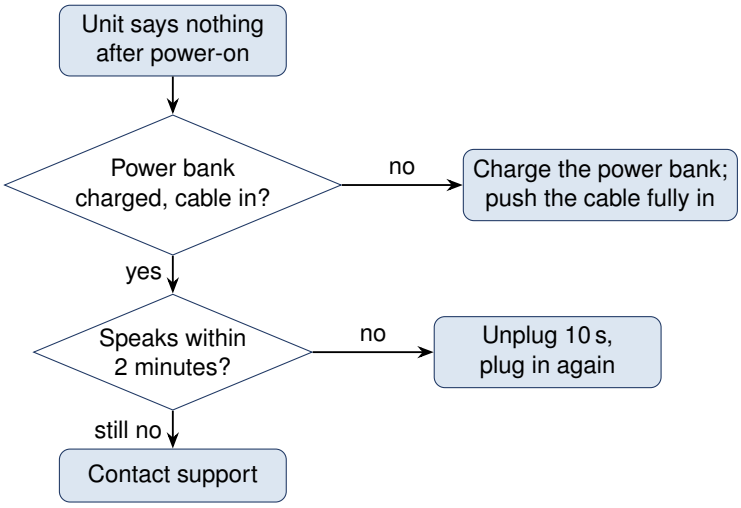


Figure 6.1: Troubleshooting flow: the unit does not start

6.2 Symptoms, Causes and Remedies

Table 6.1: Troubleshooting table

Symptom	Possible cause	Remedy
The app does not find the unit	Pairing mode not on; Bluetooth off; old pairing on the phone	Hold PAIR 5 s again; switch on Bluetooth (and Location on Android 11 or older); remove any old *DC-…* entry in Android's Bluetooth settings and pair again
Pairing box does not appear, or Pairing failed	Pairing mode closed after 2 minutes; wrong code	Hold PAIR again and type the code at once
Connected, but messages are not announced	Notification access off; app not chosen; Quiet mode on	Allow Notification access; switch the app on under Apps to announce; switch Quiet mode off
Connection drops when the phone sleeps	Battery saver stops the app	Set the app's battery use to Unrestricted
No road-sign alerts	TRIP switch at O; camera blocked; "*Sign alerts paused.*"	Set TRIP to I; clean the camera window and the windscreen; point the unit forward
No overspeed warnings	No speed-limit sign read; weak phone GPS	Normal behaviour without a sign; check the Phone GPS line is green
Alerts come late or the unit restarts	Weak power source; unit too hot	Use a power bank of 2.4 A or more; clear the vents; keep out of the sun
No trip summary when parking	Power lost with the key (car charger)	Use a power bank (Section 3.4)
Map is blank but the route shows	No mobile data on the phone	The map needs data; the route is stored anyway
The voice is too quiet or too loud	Volume setting	Change Volume in the app

6.3 Spoken Warnings

The unit has no error codes. It names the problem instead.

Table 6.2: Spoken warnings

The unit says	Meaning	Action
"*Self test: …*"	A part failed at start	Restart the unit; if repeated, contact support
"*Camera not found. Sign alerts are off.*"	Camera not detected	Restart; if repeated, contact support. Overspeed and announcements still work
"*The sign reader could not start.*"	Sign software fault	Restart; if repeated, contact support
"*Sign alerts paused.*"	Camera blocked or dazzled for 10 s	Clean the window; it resumes by itself
"*Warning: the speech stopped working.*" (or another part)	One software part stopped	It restarts by itself; if repeated, restart the unit

6.4 Removing a Pairing

To connect a different phone, or after a pairing problem:

1. On the old phone, open **Settings** → **Bluetooth** and remove (Forget) the *DC-…* device.

2. Pair the new phone as in Section 3.6.

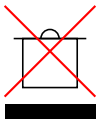
6.5 When to Contact Support

Contact support if the problem continues after the remedies above, if the case is damaged, or if you notice smoke, a burning smell or a swollen power bank (stop using it at once). Have ready: the model, the device software version (shown under the device name in the app), the app version, what the unit said, and a short description.

Chapter 7

Disposal and Environment

7.1 Do Not Dispose of with Household Waste



This symbol means the product must **not** be put in household waste. Electrical and electronic equipment contains materials that can harm the environment and health if handled incorrectly, and valuable materials that can be recovered.

7.2 How to Dispose of the Product

1. **Remove your data.** Ask a technician to erase the trips stored in the unit, or remove its microSD card. In the app, uninstall the Driver Companion app to delete the trips on the phone, and remove the *DC-…* pairing in Android's Bluetooth settings.
2. **Separate the power bank.** The unit has no internal battery. Dispose of the power bank separately at a battery collection point.
3. **Hand over** the unit and the cable to an authorised e-waste collection point or recycler: `https://ahmed12-stack.github.io/driver-companion_EPD/`.
4. Keep the receipt or recycling certificate, if one is issued.

7.3 Applicable Regulations

Table 7.1: Regulations on end-of-life handling (check the current edition and your local rules)

Regulation	Relevance
E-Waste (Management) Rules, 2022, India	Producer responsibility, collection and authorised recycling of electronic waste
Battery Waste Management Rules, 2022, India	Collection and recycling of used batteries (the power bank)
WEEE Directive 2012/19/EU	Separate collection and recovery of electrical and electronic equipment (EU)
RoHS Directive 2011/65/EU	Restriction of hazardous substances in the product

7.4 Material Information

Table 7.2: Main materials (to help recyclers)

Component	Material	Recycling route
Case and speaker pod	ABS plastic, 3D-printed	Plastics
Circuit boards (controller, amplifier, camera)	FR-4 laminate, copper, solder	E-waste
Speaker	Steel, magnet, paper cone	E-waste
Cable	Copper, PVC	E-waste
Packaging	Cardboard, paper	Paper recycling

7.5 Packaging

Recycle the packaging according to local rules. Cardboard and paper go to paper recycling.

Chapter 8

Warranty, Support and Compliance

8.1 Warranty

The DC-1 is an educational prototype and is supplied without a commercial warranty. Report any fault to the CARA team (see Technical Support below). Damage from misuse, opening the case, liquids, drops, heat, or power sources other than those in Section 3.4 is not covered.

8.2 Technical Support

Contact	Details
E-mail	`cara.support.dc1@gmail.com`
Website / downloads	`https://ahmed12-stack.github.io/driver-companion_EPD/`
Address	EPD Group 7, School of Electronics Engineering, KIIT Deemed to be University, Bhubaneswar, Odisha, India

8.3 Regulatory Compliance

Table 8.1: Compliance information

Area	Standard / regulation	Status
Product safety	IEC 62368-1 (instructional safeguards applied in this manual)	Not tested
EMC	CISPR 32; IEC 61000-4-2	Not tested
Radio	Bluetooth LE radio of the Raspberry Pi 3B+ board	Board manufacturer's own conformity
Hazardous substances	RoHS	Not assessed
Instructions for use	IEC/IEEE 82079-1	This manual

NOTICE

This product is an educational prototype and has not been certified for commercial sale.

8.4 Declaration of Conformity

Not applicable: no declaration of conformity has been issued for this prototype.

Quick Reference Card

Task	Action
Switch on	Connect the power; wait for “*System ready.*”
Start a trip	TRIP ④ to I
End a trip and hear the summary	TRIP to O
Switch off	After the summary, disconnect the power
Pair a new phone	Hold PAIR ③ for 5 s, then Pair in the app and type the spoken code
Test the speaker	App: Send test alert
Road alerts only	App: Quiet mode on
Restart	Disconnect the power for 10 s
Support	`cara.support.dc1@gmail.com`

Appendix B

Connector and Interface Details

Table B.1: Internal connections (Technician only); pin numbers on the controller's 40-pin header

Part	Pin	Signal	Description
PAIR button	11 / 9	GPIO17 / GND	Pressed = low; held 5 s = pairing mode
TRIP switch	13 / 14	GPIO27 / GND	Closed = trip running (1 s to start, 5 s to end)
Amplifier	2 / 6	5 V / GND	Supply
	12 / 35 / 40	BCLK / LRC / DIN	Digital audio (I2S)
	39	GAIN to GND	Fixed gain (12 dB)
Speaker	—	+ / −	On the amplifier's screw terminals only; never to GND
Camera	CSI	Ribbon	Camera connector, blue side towards the audio jack
Power	micro-USB	5 V DC	Controller power input

Glossary

AI remark	The one-sentence comment at the end of the trip summary, written by a small language model running inside the unit
Bluetooth LE	Bluetooth Low Energy, the short-range radio link between the phone and the unit
GPS fix	A position and speed measured by the phone's satellite receiver
IRC:67	The Indian Roads Congress code for road signs
Overspeed	Driving more than 5 km/h above the last speed limit read
Pairing	Making a trusted Bluetooth link between the phone and the unit, once
Quiet mode	An app setting that keeps road alerts and stops phone announcements
Trip	One drive, from TRIP I to TRIP O
WEEE	Waste electrical and electronic equipment
IP code	Ingress protection rating (IEC 60529)

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